Mitigating Workplace Harm through PPE-Detection

Project Group 3 - 42177 Image Processing and Pattern Recognition

1. Introduction

1.1 Background and Objectives

Workplace safety remains a persistent challenge in high-risk industries such as construction, manufacturing, and mining. In these environments, workers are routinely exposed to hazards including falling objects, moving machinery, and airborne particles. Personal protective equipment (PPE), such as hard hats, high-visibility vests, gloves, safety goggles, and masks, serves as the final line of defence when engineering and administrative controls cannot fully eliminate risk. However, PPE is only effective when it is worn correctly and consistently, and non-compliance is common due to discomfort, time pressure, or a low perception of risk.

Traditionally, PPE compliance is monitored manually by safety officers and site supervisors. This approach is labour-intensive, subjective, and difficult to scale across large or dynamic worksites where many workers move between areas throughout the day. Even where surveillance cameras are already installed, reviewing footage by hand is impractical, meaning that violations are often identified only after an incident has occurred. There is therefore a clear need for an automated, reliable, and continuous method of verifying whether workers are wearing the required safety equipment.

Recent advances in computer vision, particularly deep learning–based object detection, offer a promising solution. Two-stage detectors such as Faster R-CNN (Ren et al., 2017) and one-stage detectors such as YOLO (Redmon et al., 2016) can localise and classify multiple objects within a single image, with one-stage models achieving real-time inference speeds. Early work by Seo et al. (2015) highlighted the potential of computer vision for construction safety and health monitoring. Building on this, Fang et al. (2018) demonstrated that deep learning could detect non-hardhat use from far-field surveillance videos, while Nath et al. (2020) and Delhi et al. (2020) showed that YOLO-based models can detect multiple types of PPE on construction sites at near real-time speeds. These studies provide strong evidence that automated PPE detection is technically feasible. Nevertheless, practical challenges remain, including small object sizes, occlusion, variation in lighting and camera angles, class imbalance, and the difficulty of distinguishing between the presence and absence of equipment.

The aim of this project is to develop and evaluate an object detection system capable of identifying safety equipment in workplace images using the PPE Detection dataset. The specific objectives are to:

1. Train deep learning object detection models to recognise PPE items and cases of non-compliance.
2. Evaluate model performance using standard metrics such as precision, recall, and mean Average Precision (mAP), alongside inference speed.
3. Analyse failure cases to understand the limitations of the models under realistic conditions.
4. Assess the feasibility of deploying such a system for real-time safety monitoring.

1.2 Importance to Society

Occupational injuries and fatalities impose a heavy burden on workers, families, employers, and society as a whole. According to the International Labour Organization (ILO, 2023), approximately 2.93 million workers died from work-related causes in 2019, and more than 395 million sustained non-fatal work injuries. Sectors such as mining, construction, and utilities were identified as among the most hazardous globally based on fatal occupational injury rates. Beyond the human cost, workplace accidents result in lost productivity, medical expenses, compensation claims, legal liabilities, and project delays, all of which impose significant economic costs.

Many of these injuries are preventable. Head injuries from falling objects, eye injuries from debris, and collisions involving poorly visible workers can be substantially reduced through proper PPE use. The gap, therefore, often lies not in the availability of protective equipment but in ensuring that it is actually worn. An automated PPE detection system addresses this gap directly by providing continuous and objective monitoring. Such a system can alert supervisors to violations as they occur, enabling timely intervention before an accident happens rather than investigation after the fact.

Automated detection also offers benefits beyond immediate hazard prevention. By recording compliance data over time, organisations can identify locations, shifts, or tasks where violations frequently occur and direct training or resources accordingly. It also reduces the monitoring burden on safety personnel, allowing them to focus on higher-level risk management. Furthermore, because many worksites already operate camera infrastructure, a vision-based system can be integrated at relatively low cost. The same approach is transferable to other settings, including factories, warehouses, laboratories, and healthcare facilities, where compliance with protective equipment requirements is equally critical.

At the same time, the deployment of such technology must be approached responsibly. Continuous video analysis raises concerns about worker privacy and surveillance, and an inaccurate model may generate false alarms or overlook genuine violations. For this reason, automated PPE detection should be viewed as a tool that supports, rather than replaces, human safety management. Rigorous evaluation of model accuracy and limitations, as undertaken in this project, is therefore essential to ensure that such systems can be trusted to contribute meaningfully to safer workplaces.

2. Dataset

The Sanyal (2023) dataset contains 2,801 images at 640×640 resolution and 38,352 bounding-box annotations. Its ten classes are Person (9,872), machinery (5,346), NO-Safety Vest (4,158), Safety Cone (3,502), Hardhat (3,334), NO-Mask (3,250), Safety Vest (3,135), NO-Hardhat (2,427), Mask (1,700), and vehicle (1,628). These categories cover workers, protective equipment, missing equipment, and surrounding construction objects. The dataset therefore supports both PPE detection and broader construction scene analysis. The dataset documentation describes images collected from several sources, including videos and existing image collections, which helps explain the diversity of visual content.

Class imbalance is substantial. Person accounts for 25.7% of all annotations and has approximately six times as many examples as vehicles. Person and machinery together represent 39.7% of the boxes, while Mask and vehicle together account for only 8.7%. This concentration means that aggregate performance can hide differences between common and less frequent categories. A model could perform well on people and machinery while remaining less reliable on masks or vehicles.

The imbalance also affects comparisons between equipment presence and absence. NO-Mask has approximately 1.9 times as many annotations as Mask, while NO-Safety Vest is more frequent than Safety Vest. Hardhat, however, has more annotations than NO-Hardhat. These counts describe the annotated dataset rather than the prevalence of safety violations in real workplaces. Multiple labels can belong to the same worker, so annotation totals should not be interpreted as independent worker counts.

The split contains 2,605 training images (93.0%), 114 validation images (4.1%), and 82 test images (2.9%). Every class appears in all three splits. However, the evaluation sets are small relative to the training set, and some categories have limited annotation support. Mask has only 21 validation annotations and 28 test annotations. Consequently, a small number of missed detections could noticeably change its reported performance, making comparisons between models sensitive to individual examples.

Images often contain multiple objects of varying sizes, including small PPE items and large machinery. Visual samples show differences in lighting, viewpoint, and crowding, as well as precomposed mosaics. These characteristics introduce variation beyond class frequency alone. Detecting a distant helmet or partially visible mask presents a different challenge from recognizing a large excavator. The mosaics also combine different scenes within individual images, which distinguishes parts of the training data from ordinary construction photographs.

Integrity checks found no corrupt images, missing label files, invalid class IDs, or invalid bounding-box coordinates. There are 24 empty-label images and two duplicate annotation rows. Empty-label images may serve as background examples, although their annotation completeness requires visual confirmation. No exact image duplicates were found using file and decoded-pixel hashes. However, eight filename-derived source groups overlap between training and evaluation splits. This suggests possible leakage from related source images, even though the image files themselves are different.

The dataset uses normalized YOLO bounding-box annotations and is compatible with YOLO26 detection. Its existing classes and annotations can therefore be reused for a YOLO26 PPE detection experiment. The class mapping includes missing-equipment categories, allowing experiments to distinguish annotated equipment absence from the presence of protective items.

3. Methods

3.1 Data Challenges and Mitigations

The data characteristics described in Section 2 motivate a detection pipeline that addresses class imbalance, variation in object size, visual ambiguity and possible source leakage. These challenges influence both model selection and training design: detecting a large vehicle does not establish that a model can recognise a distant mask or an uncovered head. The proposed comparison therefore considers Faster R-CNN, SSD, YOLO26 and RT-DETR under the common evaluation framework in Section 4, with particular attention to the six PPE and missing-equipment classes.

Class imbalance requires attention because Person has 9,872 annotations, whereas Mask and vehicle have only 1,700 and 1,628, respectively. Johnson and Khoshgoftaar (2019) review sampling and cost-sensitive learning as approaches to imbalanced deep learning. For this project, the initial comparison should retain each detector's standard objective and use identical training images. If minority-class recall remains weak, a separate ablation can test modest oversampling of training images containing rare PPE classes. Sampling must operate at image level because each image can contain several categories; repeatedly sampling masks can also increase exposure to people and other objects. Per-class AP and violation recall will determine whether the intervention helps.

Foreground and background imbalance is a separate problem: dense detectors evaluate many locations that contain no labelled object. Lin et al. (2017) introduced focal loss to reduce the contribution of well-classified examples and concentrate learning on difficult cases. This motivates attention to each architecture's mechanism for controlling easy background examples. However, focal loss should not be described as part of every detector or substituted into SSD's baseline objective. Any change to a loss function should be evaluated separately so that architectural differences remain distinguishable from training interventions.

Object size varies from small helmets and masks to large machinery. The proposed baseline preserves the dataset's 640×640 resolution, subject to architecture compatibility, and uses detectors with predictions at multiple spatial scales. Resizing must transform bounding boxes consistently and avoid inadvertently discarding tiny annotations. Scale jitter can expose models to different apparent viewing distances, but aggressive downscaling may remove the visual evidence needed to recognise PPE. Small-object AP and manually reviewed misses will therefore be used to assess whether the chosen resolution and augmentation settings are adequate. Increasing pixel dimensions alone cannot recover detail absent from the original image.

Lighting, viewpoint and partial occlusion motivate horizontal flips, moderate colour jitter and controlled scale transformations applied only during training. Shorten and Khoshgoftaar

(2019) explain how augmentation can expand visual variation and reduce overfitting when labelled data are limited. Here, transformed boxes must remain valid, and crops must preserve enough evidence for the assigned class. Additional mosaic augmentation should be tested cautiously because the dataset already contains precomposed mosaics. The baseline should use a shared augmentation family, while validation and test images receive deterministic preprocessing.

Annotation semantics also constrain interpretation. NO-Hardhat, NO-Mask and NO-Safety Vest are explicitly annotated categories; a failure to detect equipment does not itself establish a violation. An occluded face or head may provide insufficient evidence either way. The proposed error analysis will therefore inspect confusion between equipment-present and equipment-absent labels, especially where boxes overlap on the same worker. The two duplicate annotation rows should be removed, and the 24 empty-label images should be visually checked before being retained as background examples. These checks address label quality without assuming that every empty annotation file is correct.

Finally, the eight filename-derived source groups spanning training and evaluation require investigation. Kapoor and Narayanan (2023) show that leakage can produce misleading estimates of generalisation. Source overlap is a warning rather than proof of leakage, so related images should first be inspected. If common videos, scenes or source photographs are confirmed, an additional group-separated evaluation should keep each source within one split. The supplied split can remain the common benchmark, with its limitation documented. The small validation and test sets also motivate restrained hyperparameter tuning, repeated training seeds and uncertainty intervals, with all thresholds selected before inspecting test results.

3.2 Techniques

In this experiment, we compare classical object detection families with key modern architectural families. Faster R-CNN provides an established two-stage, proposal-based reference, while SSD provides an established one-stage detector against which advances in newer real-time architectures can be assessed. YOLO26 represents a contemporary deployment-oriented detector designed around a strong accuracy–latency trade-off, whereas RT-DETR provides an architecturally distinct transformer-based, end-to-end alternative. The comparison therefore examines how each architecture performs on the specific challenges of PPE detection.

3.2.1 SSD

Single Shot MultiBox Detector (SSD) is included as an established one-stage, anchor-based baseline. Liu et al. (2016) describe a network that predicts object categories and bounding-box adjustments directly, without a separate proposal stage. Default boxes of different scales and aspect ratios are placed across several feature maps, enabling predictions for objects of different sizes. The original architecture uses a VGG-16 base with additional convolutional layers. Its training combines classification and localisation losses, matches default boxes to annotated objects, and uses hard negative mining to limit background examples relative to positives. At inference, non-maximum suppression removes redundant predictions (Liu et al., 2016).

For this dataset, SSD offers a concrete test of whether an established detector can provide adequate PPE recall within the project's latency target. Its inclusion adds a different one-stage design to the comparison, helping the team examine whether improvements from newer models justify their deployment cost. The comparison will measure that trade-off on the same hardware and test images rather than transfer published benchmark speeds to this application. Particular attention will be given to the relationship between overall mAP and recall for missing-equipment categories, because a fast detector is useful only if its alerts are sufficiently reliable.

The proposed implementation starts from the default Imagenet-pretrained SSD300 checkpoint, replaces its classification outputs with the ten dataset classes plus background, and fine-tunes the detector using the common training budget. YOLO annotations will be converted to the coordinate convention required by the implementation, with explicit checks of class indices and box positions after resizing. The exact backbone, checkpoint and default-box configuration will be recorded. While other models can accept a 640x640 input, the SSD300 model variant accepts a 300x300 image, so this will be reported as a protocol difference.

The experiment will examine default-box coverage for small PPE and the frequency of missed masks and helmets. If these objects rarely receive suitable matches, changing default-box scales can be investigated in a separate ablation while retaining the original configuration as the baseline. Confidence and suppression thresholds will be chosen on validation data and then fixed for testing. Speed measurements will include resizing, network inference and suppression at batch size one. SSD will be judged using the same per-class AP, violation precision and recall, and latency criteria as the other detectors, allowing its suitability to follow from measured performance rather than its architectural simplicity.

3.2.2 Faster R-CNN

Faster R-CNN represents an established two-stage, proposal-based detector. The architecture introduces a Region Proposal Network (RPN) that shares convolutional features with the detection network. The RPN predicts candidate object locations and objectness scores, after which candidate regions are classified and their bounding boxes refined by the second-stage detection head (Ren et al., 2017). This differs fundamentally from the dense single-stage prediction used by SSD and YOLO and therefore provides a useful reference for whether the additional proposal and refinement stage improves detection quality sufficiently to justify its computational cost. Ren et al. demonstrated that sharing features between proposal generation and detection substantially reduced the proposal-generation overhead of earlier region-based detectors.

Faster R-CNN has been used for PPE detection in construction-safety monitoring. Fang et al. (2018) applied Faster R-CNN to non-hardhat-use detection using far-field surveillance footage collected from 25 construction sites. Their evaluation specifically considered variations in illumination, background, worker posture and viewing conditions and reported precision and recall above 90% across the evaluated conditions. However, Wang (2025) also identified its two-stage processing as a disadvantage for real-time deployment compared to single-stage detectors.

In our study, Faster R-CNN therefore tests whether proposal-based detection and subsequent region refinement provide measurable benefits for difficult PPE instances, including small, partially occluded or visually ambiguous objects. The comparison will examine per-class AP, small-object AP and violation recall alongside inference latency. The model will be initialised from its default ImageNet pretrained checkpoint, with its classification head adapted to the ten dataset classes plus background and fine-tuned using the common data split and training budget. The exact backbone, proposal configuration and checkpoint will be recorded, and latency measurements will include proposal generation, detection and post-processing.

3.2.3 YOLO 26

YOLO26 is included as the contemporary, deployment-oriented one-stage detector. YOLO architectures have been widely applied to PPE detection because they perform localisation and classification in a single integrated pass. Nath et al. (2020) used YOLO-based models for simultaneous worker and PPE detection, while Wang et al. (2021) found a clear capacity–efficiency trade-off among YOLOv5 variants, with YOLOv5x achieving the highest mAP and YOLOv5s the fastest inference in their PPE experiments.

YOLO26 changes several components of earlier YOLO generations. It removes Distribution Focal Loss from bounding-box regression, simplifying the detection head, and adds an optional one-to-one prediction head that produces final detections without NMS, alongside the default one-to-many head, which retains NMS (Jocher et al., 2026). Its training recipe combines the MuSGD optimiser, Progressive Loss and Small-Target-Aware Label Assignment (STAL), the last of which is intended to maintain positive training assignments for small objects (Jocher et al., 2026).

The proposed implementation fine-tunes COCO-pretrained YOLO26-M weights, with the detection head resized to the ten dataset classes. At 640×640 input, YOLO26-M has approximately 20.4 million parameters and achieves 53.1% mAP@[0.5:0.95] on COCO val2017 (Ultralytics, 2026a) which is a comparable AP compared to RT-DETR-L (53.0%). The default one-to-many head with NMS serves as the primary configuration. Finally, latency is measured end to end at batch size one, including pre-processing and NMS where applied.

3.2.4 RT-DETR

RT-DETR is included as the transformer-based, end-to-end representative in the comparison. It adapts DETR-style detection to real-time use. Features from a convolutional backbone pass to a hybrid encoder, in which self-attention is applied only to the highest-level feature map and the remaining scales are combined through convolutional fusion. A fixed set of object queries, initialised from encoder features by IoU-aware selection, is refined by a deformable-attention decoder, with each query producing one box and class prediction (Zhao et al., 2024). Training assigns predictions to ground-truth objects by one-to-one (Hungarian) matching, so the network outputs final detections directly and requires no non-maximum suppression (NMS) (Zhao et al., 2024).

For this dataset, RT-DETR tests whether an attention-based design offers advantages on the challenges as described earlier. Global attention may help separate Hardhat from NO-Hardhat, where the surrounding head and body provide context, and the absence of

NMS removes the risk of true detections being suppressed among overlapping workers. Small PPE items remain a concern, because their detection depends on the convolutional fusion path and the decoder's multi-scale sampling. Contrasting RT-DETR with YOLO26, brings the attention-based design into focus, whereas contrasting it with Faster R-CNN and SSD reflects the combined influence of architecture and post-processing.

The proposed implementation fine-tunes COCO-pretrained RT-DETR-L weights, which pairs the hybrid encoder and decoder with an HGNetv2 backbone (32 M parameters, 53.0% AP on COCO; Ultralytics, 2026b), with the output head resized to the ten dataset classes. The model variant chosen has roughly the same AP performance on the pretrained COCO dataset as the YOLO-26m(53.1% AP) model. The native 640×640 resolution is retained. Latency is measured end to end at batch size one, including pre-processing and output decoding, even though no suppression step is applied.

4. Evaluation

4.1 Evaluation Metrics

Models are assessed on three axes: detection accuracy, operating-point reliability and inference speed. These serve objectives (2) and (4), and a structured error analysis serves objective (3).

Detection matching. A predicted box is a true positive (TP) if its class matches a ground-truth box and their Intersection over Union (IoU) reaches a threshold. Each ground-truth box can be matched only once, so further overlapping predictions count as false positives (FP), and unmatched ground-truth boxes are false negatives (FN).

Precision, recall and F1. Precision (TP / (TP + FP)) is the share of alerts that are correct, and recall (TP / (TP + FN)) is the share of real objects that are found. In a safety setting the two errors carry different costs: low recall means missed violations, while low precision means false alarms that erode supervisors' trust. We therefore report both, together with F1, per class at a single confidence threshold chosen on the validation set (the value that maximises F1).

Average precision and mAP. Precision and recall trade off as the confidence threshold varies, so each class is summarised by its average precision (AP), the area under its precision–recall curve, and AP is averaged across classes to give mAP. We report two variants. mAP@0.5 (IoU ≥ 0.5; Everingham et al., 2010) tolerates imprecise boxes and suffices to decide whether an item is present. mAP@[0.5:0.95] (Lin et al., 2014) averages over ten IoU thresholds from 0.5 to 0.95 and rewards tight localisation. Each is computed over all ten classes and over the six PPE classes (Hardhat, Mask, Safety Vest and their NO-counterparts), since Person, Safety Cone, machinery and vehicle are context rather than the project's target. Because mAP is macro-averaged, rare classes count as much as common ones. Per-class AP is also reported so that a strong average cannot hide a weak violation class such as NO-Hardhat, and AP is broken down by object size (small, medium, large) to expose the small-object weakness noted in the Introduction.

Inference speed. We report latency (ms per image) and throughput (frames per second, FPS) at batch size 1, averaged over [200] images after warm-up on identical hardware,

including pre-processing and non-maximum suppression (NMS) where applicable, since a deployed system pays for both. Parameter count is reported as a proxy for deployability.

Error analysis. A per-class confusion matrix (with a background row and column for spurious and missed detections) locates systematic confusions, such as Hardhat versus NO-Hardhat. We then manually review the highest-confidence false positives and the missed objects, grouping them by cause: occlusion, small size, lighting, and unusual pose or camera angle.

4.2 Experimental Setup

Controlled comparison. The models described in Section 3 are trained under matched conditions: identical splits, the same input resolution [640 px], the same augmentation family [horizontal flip, colour jitter, scale jitter] and the same training budget [100+ epochs] with early stopping on validation mAP@0.5. Architecture-specific settings follow each model's recommended defaults, and the learning rate is tuned over a small grid on the validation set with equal effort per model, so that differences reflect the architectures rather than unequal tuning.

Unified evaluation. YOLO-format labels are converted once to COCO JSON, and all predictions are scored against them by a single evaluator [e.g. pycocotools] rather than each framework's built-in validation routine, whose AP interpolation, NMS and confidence filtering differ subtly. For AP computation, predictions are kept down to a low confidence [0.001], whereas precision, recall and F1 use the validation-selected threshold.

Results are compared with published PPE detectors (Nath et al., 2020; Delhi et al., 2020) only indicatively, since those studies used their own datasets and protocols.

4.3 Definition of Success

Success criteria are fixed before test-set results are examined, so that thresholds cannot be adjusted afterwards. The values below are proposals for the team to confirm.

1. Accuracy: mAP@0.5 ≥ [0.75] over the six PPE classes on the test set, with no PPE class below [0.50] AP@0.5.
2. Violation reliability: for NO-Hardhat, NO-Mask and NO-Safety Vest, recall ≥ [0.85] and precision ≥ [0.80] at the chosen threshold. Recall is weighted higher because a missed violation defeats the system's purpose, while the precision floor keeps alerts actionable.
3. Real-time capability: at least [25] FPS at batch size 1 on the reference hardware, roughly the frame rate of standard video.
4. Model comparison: a model is preferred if it meets criteria 1–3. Where several models do, differences smaller than the seed-to-seed standard deviation or inside the bootstrap interval are treated as comparable, and the faster or smaller model is preferred.

Criteria are judged on test-set point estimates, with intervals reported so that a narrow pass or miss is read with caution.

Meeting these targets is not the only measure of success. The project also succeeds if it documents the main failure modes and gives an evidence-based judgement on deployment feasibility (objectives 3 and 4). A model that misses a target is a legitimate finding, provided the cause is explained. Consistent with the Introduction, results are interpreted as evidence for a decision-support tool that assists, rather than replaces, human safety management.

5. Assumptions and Limitations

We acknowledge several assumptions and limitations of the experimental setup. First, Faster R-CNN and SSD are initialised from ImageNet-pretrained backbones, whereas YOLO26-M and RT-DETR-L are initialised from COCO-pretrained detector weights. These checkpoints were chosen on the basis of availability in Matlab and Ultralytics, as well as estimated effort of integration. As a result, performance differences between the established and contemporary models reflect both architecture and pretraining.

Intra-family ablation studies can also be performed across different YOLO26 and RT-DETR model sizes to further understand accuracy and latency performance across and within either family. However this would result in an extended delivery timeline well beyond what's available to this course, and are therefore left as future work.

6. Estimated Delivery Timeline

The estimated delivery timeline is shown in the table below. After initial data cleaning and evaluation harness setup, model implementation, and training can be executed in parallel. Buffers were also set to account for unexpected delays.

Task	Estimate (Days)
Environment setup and data cleaning	2
Label conversion and unified evaluation pipeline	2
Model training: Faster R-CNN, SSD, YOLO26, RT-DETR (parallel)	5
Test-set evaluation and latency benchmarking	1
Error analysis	2
Report writing and proofreading	2
Buffer for unexpected delays	4
Total	**18**

7. Conclusion

PPE compliance is an important component of workplace safety because protective equipment is only effective when it is worn correctly and consistently. Manual monitoring is difficult to sustain across large and dynamic worksites, creating a need for automated methods that can identify non-compliance early enough for supervisors to intervene before an incident occurs. This project therefore frames PPE compliance monitoring as an object detection problem, where a model must localise workers and safety equipment while also recognising explicitly annotated cases of missing equipment.

The proposed system will be developed using the PPE Detection dataset, which contains 2,801 images and 38,352 bounding-box annotations across ten classes. Importantly, the dataset includes both PPE-present and PPE-absent classes, such as Hardhat and NO-Hardhat, allowing the task to represent compliance violations directly rather than inferring non-compliance solely from a failure to detect equipment. The dataset also presents practical challenges including class imbalance, small PPE objects, occlusion, varying viewpoints and lighting, and possible source overlap between splits.

To investigate how different detection paradigms address these challenges, the study compares Faster R-CNN, SSD, YOLO26 and RT-DETR. Faster R-CNN and SSD provide established two-stage and one-stage references, while YOLO26 and RT-DETR represent more recent real-time and end-to-end approaches.

Evaluation is designed around the underlying safety objective rather than accuracy alone. mAP and per-class AP measure overall detection capability, while precision and recall indicate the practical consequences of false alarms and missed violations. Particular attention is given to the missing-equipment classes and small objects, while latency and FPS assess whether detection can operate quickly enough for real-time monitoring. Error analysis will further identify conditions under which violations are missed or incorrectly reported. Ultimately, the aim is to identify whether any of the evaluated models provides a sufficiently accurate, reliable and efficient basis for a real-time PPE compliance system that supports human safety management.

8. References

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9. Team Member Contributions

Student Name	Student ID	Contribution Weight	Contribution Detail
Harunori Kawano	25741099	1	Authored Introduction: Investigated the background of the issue and summarized the details
Eunkwang Shin	26170530	1	Authored Dataset part with exploratory data analysis.
Van An Hoang	26046146	1	Authored Evaluation: Propose framework for experimentation and metrics for evaluation among models for fairness and comparability.
Bhargav Ram Anantla	25619875	1	Authored data challenges and motivation for techniques, and SSD.
Leiko Armand Ravelo	27045171	1	Authored Faster R-CNN and YOLO26, proofreading
Siddhant Salgia	26025155	1	Authored RT-DETR, proofreading